



《数值分析》7

主要内容:

大型稀疏矩阵的背景

Jacobi迭代与Seidel迭代

迭代法的矩阵表示

迭代法数值实验

大型稀疏矩阵的背景（一维情况）



二阶常微分方程：
（两点边值问题）

$$\begin{cases} y'' + y + x = 0, & x \in (0,1) \\ y(0) = 0, y(1) = 0. \end{cases}$$

举例：在某区域内求流体的速度或静电场的电位，当这区域边界上的速度或电位已经知道时

令 $h = 1/(n+1)$, $x_j = jh$, $y_j = y(x_j)$ ($j = 0, 1, \dots, n+1$)

$$\frac{y_{j+1} - 2y_j + y_{j-1}}{h^2} + y_j + x_j = 0 \quad (j = 1, 2, \dots, n)$$

三对角方程组 $-y_{j-1} + (2 - h^2)y_j - y_{j+1} = x_j h^2$

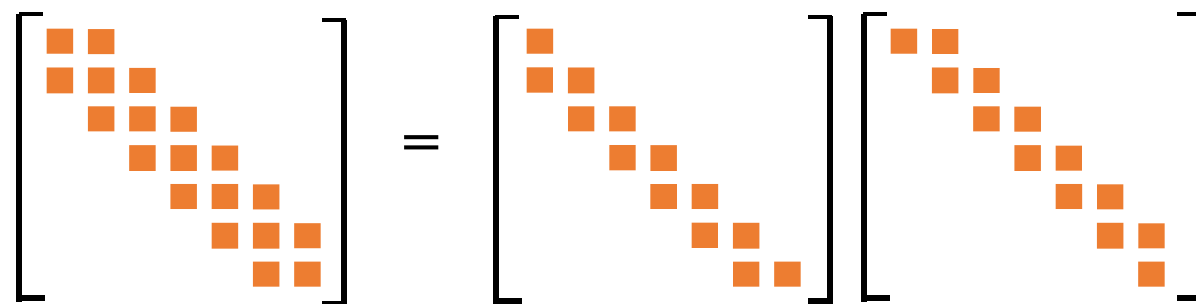
$$\begin{bmatrix} 2-h^2 & -1 & & & \\ -1 & 2-h^2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2-h^2 & -1 \\ & & & -1 & 2-h^2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_{n-1} \\ y_n \end{bmatrix} = h^2 \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix}$$

一般形式

$$\begin{bmatrix} b_1 & c_1 & & & \\ a_2 & b_2 & c_2 & & \\ & \ddots & \ddots & \ddots & \\ & & a_{n-1} & b_{n-1} & c_{n-1} \\ & & & a_n & b_n \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} = \begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_{n-1} \\ f_n \end{bmatrix}$$

三角分解:

$$A = LU$$



三对角矩阵

单位下三角阵

上三角阵

$$AX=F \rightarrow \underline{L}UX=F$$

$$\textcircled{1} LY=F \quad \textcircled{2} UX=Y$$

④ $y'' + y + x = 0, x \in (0, 1)$
 $y(0) = 0, y(1) = 0$
 $y_j + y_{j+1} + x_j = 0$
 $\rightarrow \frac{y_{j+1} - 2y_j + y_{j-1}}{h^2} + y_j + x_j = 0$
 $\rightarrow y_{j+1} + (h^2 - 2)y_j + y_{j-1} = -h^2 x_j$

$j=1, (h^2 - 2)y_1 + y_2 = -h^2 x_1 - y_0$
 $j=2, y_1 + (h^2 - 2)y_2 + y_3 = -h^2 x_2$
 $\vdots$
 $j=n, y_{n-1} + (h^2 - 2)y_n = -h^2 x_n - y_{n+1}$

$$A y = f$$

$y = f(x), x \in \mathbb{R}$



$x \rightarrow \begin{array}{c} 0 \quad h \quad 1 \\ x_0 \quad x_1 \quad x_2 \quad \dots \quad x_n \quad x_{n+1} \end{array}$

① $h = \frac{1}{n+1}, x_j = jh (j=0, \dots, n+1)$

② $y' = f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} = \lim_{n \rightarrow \infty} \frac{f(x_{j+1}) - f(x_j)}{h}$

$y'_{j+1} \approx \frac{y_{j+1} - y_j}{h}$ (向前差分)
 $y'_j \approx \frac{y_j - y_{j-1}}{h}$ (向后差分)

② $y'' = f''(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$y''_j = \frac{y'_{j+1} - y'_j}{h} = \frac{\frac{y_{j+1} - y_j}{h} - \frac{y_j - y_{j-1}}{h}}{h} = \frac{y_{j+1} - 2y_j + y_{j-1}}{h^2}$

大型稀疏矩阵的背景 (二维情况)

边值问题:
$$\begin{cases} u_{xx} + u_{yy} = 0, & 0 < x, y < 1 \\ u(0, y) = u(x, 0) = u(x, 1) = 0 \\ u(1, y) = \sin \pi y \end{cases}$$

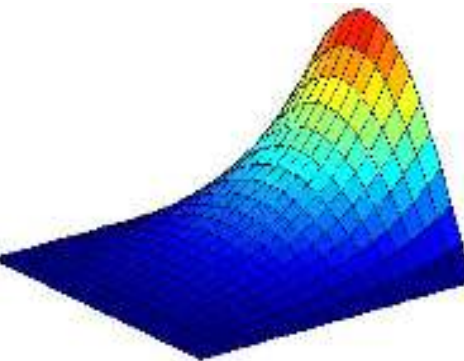
令 $h = 1/(n+1)$, $x_i = ih$, $y_j = jh$ ($i, j = 0, 1, \dots, n+1$)

记 $u_{i,j} = u(x_i, y_j)$, ($i, j = 0, 1, \dots, n+1$)

$$\frac{u_{i+1,j} - 2u_{ij} + u_{i-1,j}}{h^2} + \frac{u_{i,j+1} - 2u_{ij} + u_{i,j-1}}{h^2} = 0$$

$$u_{i,j-1} + u_{i-1,j} - 4u_{ij} + u_{i+1,j} + u_{i,j+1} = 0$$

$$\Rightarrow \begin{bmatrix} u_{1,j-1} \\ u_{2,j-1} \\ u_{3,j-1} \\ u_{4,j-1} \end{bmatrix} + \begin{bmatrix} u_{0,j} \\ u_{1,j} \\ u_{2,j} \\ u_{3,j} \end{bmatrix} - 4 \begin{bmatrix} u_{1,j} \\ u_{2,j} \\ u_{3,j} \\ u_{4,j} \end{bmatrix} + \begin{bmatrix} u_{2,j} \\ u_{3,j} \\ u_{4,j} \\ u_{5,j} \end{bmatrix} + \begin{bmatrix} u_{1,j+1} \\ u_{2,j+1} \\ u_{3,j+1} \\ u_{4,j+1} \end{bmatrix} = 0$$



($n=4$)

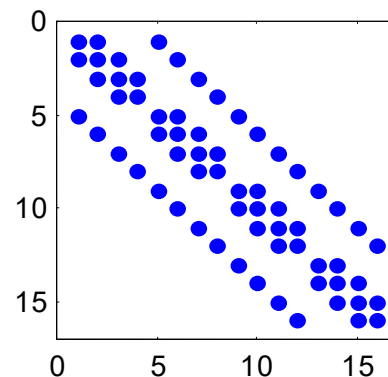
大型稀疏矩阵的背景（二维情况）

$$U_1 = \begin{bmatrix} u_{11} \\ u_{21} \\ u_{31} \\ u_{41} \end{bmatrix} \quad U_2 = \begin{bmatrix} u_{12} \\ u_{22} \\ u_{32} \\ u_{42} \end{bmatrix} \quad U_3 = \begin{bmatrix} u_{13} \\ u_{23} \\ u_{33} \\ u_{43} \end{bmatrix} \quad U_4 = \begin{bmatrix} u_{14} \\ u_{24} \\ u_{34} \\ u_{44} \end{bmatrix} \quad \Rightarrow \quad U = \begin{bmatrix} U_1 \\ U_2 \\ U_3 \\ U_4 \end{bmatrix}$$

$$\begin{cases} BU_1 + U_2 = F_1 \\ U_1 + BU_2 + U_3 = F_2 \\ U_2 + BU_3 + U_4 = F_3 \\ U_3 + BU_4 = F_4 \end{cases}$$

$$B = \begin{bmatrix} -4 & 1 & & \\ 1 & -4 & 1 & \\ & 1 & -4 & 1 \\ & & 1 & -4 \end{bmatrix}$$

$$AU = F \quad A = \begin{bmatrix} B & I & & \\ I & B & I & \\ & I & B & I \\ & & I & B \end{bmatrix}$$



线性系统迭代法引例

例4.1
$$\begin{cases} 9x_1 - x_2 - x_3 = 7 \\ -x_1 + 10x_2 - x_3 = 8 \\ -x_1 - x_2 + 15x_3 = 13 \end{cases}$$

特点: 系数矩阵主对角元均不为零

$$\Leftrightarrow \begin{cases} x_1 = (7 + x_2 + x_3) / 9 \\ x_2 = (8 + x_1 + x_3) / 10 \\ x_3 = (13 + x_1 + x_2) / 15 \end{cases} \quad \text{取 } X^{(0)} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

计算格式 $X^{(1)} = B X^{(0)} + f$

$$\begin{bmatrix} x_1^{(1)} \\ x_2^{(1)} \\ x_3^{(1)} \end{bmatrix} = \begin{bmatrix} 0 & 1/9 & 1/9 \\ 1/10 & 0 & 1/10 \\ 1/15 & 1/15 & 0 \end{bmatrix} \begin{bmatrix} x_1^{(0)} \\ x_2^{(0)} \\ x_3^{(0)} \end{bmatrix} + \begin{bmatrix} 7/9 \\ 8/10 \\ 13/15 \end{bmatrix}$$



线性系统迭代法引例

计算格式: $X^{(k+1)}=BX^{(k)}+f$

$X^{(0)}$	$X^{(1)}$	$X^{(2)}$	$X^{(3)}$	$X^{(4)}$	
0	0.7778	0.9630	0.9929	0.9987	
0	0.8000	0.9644	0.9935	0.9988	
0	0.8667	0.9778	0.9952	0.9991	

准确解 $\rightarrow$

X^*
1.0000
1.0000
1.0000

雅可比迭代法

[illegible]

$$\mathbf{x}_i^{(k+1)} = \frac{1}{a_{ii}} [b_i - \sum_{j=1}^{i-1} a_{ij} \mathbf{x}_j^{(k)} - \sum_{j=i+1}^n a_{ij} \mathbf{x}_j^{(k)}]$$

$$(i = 1, 2, \dots, n; k = 1, 2, \dots)$$

取初始向量 $X^{(0)}=[x_1^{(0)} \ x_2^{(0)} \ \cdots \ x_n^{(0)}]^T$, 迭代计算

上一页例子即为
Jacobi 迭代

迭代法适用于解**大型稀疏方程组**

(万阶以上的方程组, 系数矩阵中零元素占很大比例, 而非零元按某种模式分布)

背景: 电路分析、边值问题的数值解和数学物理方程

问题: (1)如何构造迭代格式?

(2)迭代格式是否收敛?

(3)收敛速度如何?

(4)如何进行误差估计?

对比:
迭代法方程求根的迭代法

高斯-赛德尔迭代法

$$\sum_{j=1}^n a_{ij} x_j = b_i \quad (i = 1, 2, \dots, n)$$

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left[b_i - \sum_{j=1}^{i-1} a_{ij} x_j^{(k+1)} - \sum_{j=i+1}^n a_{ij} x_j^{(k)} \right]$$

$$(i = 1, 2, \dots, n; k = 1, 2, \dots)$$

取初始向量 $x^{(0)} = [x_1^{(0)} \ x_2^{(0)} \ \cdots \ x_n^{(0)}]^T$, 迭代计算

例
$$\begin{cases} 9x_1 - x_2 - x_3 = 7 \\ -x_1 + 10x_2 - x_3 = 8 \\ -x_1 - x_2 + 15x_3 = 13 \end{cases} \iff \begin{cases} x_1 = (7 + x_2 + x_3)/9 \\ x_2 = (8 + x_1 + x_3)/10 \\ x_3 = (13 + x_1 + x_2)/15 \end{cases}$$

$$x_1^{(k+1)} = (7 + x_2^{(k)} + x_3^{(k)})/9$$

$$x_2^{(k+1)} = (8 + x_1^{(k+1)} + x_3^{(k)})/10$$

$$x_3^{(k+1)} = (13 + x_1^{(k+1)} + x_2^{(k+1)})/15$$

$$\begin{bmatrix} x_1^{(0)} \\ x_2^{(0)} \\ x_3^{(0)} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ -1/10 & 1 & 0 \\ -1/15 & -1/15 & 1 \end{bmatrix} \begin{bmatrix} x_1^{(k+1)} \\ x_2^{(k+1)} \\ x_3^{(k+1)} \end{bmatrix} = \begin{bmatrix} 0 & 1/9 & 1/9 \\ 0 & 0 & 1/10 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1^{(k)} \\ x_2^{(k)} \\ x_3^{(k)} \end{bmatrix} + \begin{bmatrix} 7/9 \\ 8/10 \\ 13/15 \end{bmatrix}$$

迭代法解线性方程组

$$\begin{cases} 9x_1 - x_2 - x_3 = 7 \\ -x_1 + 10x_2 - x_3 = 8 \\ -x_1 - x_2 + 15x_3 = 13 \end{cases}$$

雅可比迭代法实验数据

0.7778	0.8000	0.8667
0.9630	0.9644	0.9719
0.9929	0.9935	0.9952
0.9987	0.9988	0.9991
0.9998	0.9998	0.9998
1.0000	1.0000	1.0000

赛德尔迭代法实验数据

0.7778	0.8778	0.9770
0.9839	0.9961	0.9987
0.9994	0.9998	0.9999
1.0000	1.0000	1.0000
1.0000	1.0000	1.0000

总结：雅可比迭代法的矩阵表示

将方程组 $AX = b$ 的系数矩阵 A 分解

$$A = D - U - L$$

$$D = \begin{bmatrix} a_{11} & & & \\ & a_{22} & & \\ & & \ddots & \\ & & & a_{nn} \end{bmatrix}$$

$$L = \begin{bmatrix} 0 & & & \\ a_{21} & 0 & & \\ \vdots & \ddots & \ddots & \\ a_{n1} & \cdots & a_{n,n-1} & 0 \end{bmatrix}$$

$$U = \begin{bmatrix} 0 & a_{12} & \cdots & a_{1n} \\ & \ddots & \ddots & \vdots \\ & & 0 & a_{n-1,n} \\ & & & 0 \end{bmatrix}$$

$$AX = b \Rightarrow DX^{(k+1)} = (U+L)X^{(k)} + b$$

$$X^{(k+1)} = D^{-1}(U+L)X^{(k)} + D^{-1}b$$

$$\text{记 } B_J = D^{-1}(U+L) \quad X^{(k+1)} = B_J X^{(k)} + f_J$$

雅可比迭代矩阵

$$B_J = \begin{bmatrix} a_{11} & & & \\ & a_{22} & & \\ & & \ddots & \\ & & & a_{nn} \end{bmatrix}^{-1} \begin{bmatrix} 0 & -a_{12} & \cdots & -a_{1n} \\ -a_{21} & 0 & \cdots & -a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ -a_{n1} & -a_{n2} & \cdots & 0 \end{bmatrix}$$

$$B_J = \begin{bmatrix} 0 & -a_{12}/a_{11} & \cdots & -a_{1n}/a_{11} \\ -a_{21}/a_{22} & 0 & \cdots & -a_{2n}/a_{22} \\ \cdots & \cdots & \cdots & \cdots \\ -a_{n1}/a_{nn} & -a_{n2}/a_{nn} & \cdots & 0 \end{bmatrix} \quad f_J = \begin{bmatrix} b_1/a_{11} \\ b_2/a_{22} \\ \vdots \\ b_n/a_{nn} \end{bmatrix}$$

高斯-赛德尔迭代法的矩阵表示

$$a_{ii}x_i^{(k+1)} = [b_i - \sum_{j=1}^{i-1} a_{ij}x_j^{(k+1)} - \sum_{j=i+1}^n a_{ij}x_j^{(k)}]$$

$$\sum_{j=1}^i a_{ij}x_j^{(k+1)} = b_i - \sum_{j=i+1}^n a_{ij}x_j^{(k)} \quad (i = 1, 2, \dots, n)$$

$$\begin{bmatrix} a_{11} & & & \\ a_{21} & a_{22} & & \\ \vdots & \vdots & \ddots & \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1^{(k+1)} \\ x_2^{(k+1)} \\ \vdots \\ x_n^{(k+1)} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} - \begin{bmatrix} 0 & a_{12} & \cdots & a_{1n} \\ & 0 & \ddots & \vdots \\ & & \ddots & a_{n-1,n} \\ & & & 0 \end{bmatrix} \begin{bmatrix} x_1^{(k)} \\ x_2^{(k)} \\ \vdots \\ x_n^{(k)} \end{bmatrix}$$

$$(D - L)X^{(k+1)} = b + UX^{(k)}$$

$$X^{(k+1)} = (D - L)^{-1}b + (D - L)^{-1}UX^{(k)}$$

记 $B_{G-S}=(D-L)^{-1}U$, $f_{G-S}=(D-L)^{-1}b$

高斯-赛德尔迭代格式: $X^{(k+1)}=B_{G-S}X^{(k)}+f_{G-S}$

$$B_{G-S} = \begin{bmatrix} a_{11} & & & \\ a_{21} & a_{22} & & \\ \vdots & \vdots & \ddots & \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}^{-1} \begin{bmatrix} 0 & a_{12} & \cdots & a_{1n} \\ & 0 & \ddots & \vdots \\ & & \ddots & a_{n-1,n} \\ & & & 0 \end{bmatrix}$$

$$f_{G-S} = \begin{bmatrix} a_{11} & & & \\ a_{21} & a_{22} & & \\ \vdots & \vdots & \ddots & \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}^{-1} \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

总结： 矩阵分裂导出的迭代法

$$A = M - N \quad (\text{要求 } M \text{ 为可逆矩阵})$$

$$AX = b \rightarrow (M - N)X = b \rightarrow MX = NX + b$$

$$\rightarrow X^{(k+1)} = (M^{-1}N) X^{(k)} + M^{-1}b$$

取 $M = D \rightarrow$ 雅可比迭代法

$$A = D - (D - A) \rightarrow$$

$$X^{(k+1)} = D^{-1}[(D - A) X^{(k)} + b] \rightarrow$$

$$X^{(k+1)} = X^{(k)} + D^{-1}[b - AX^{(k)}]$$

$$\text{记 } r_k = b - AX^{(k)} \quad \rightarrow \quad X^{(k+1)} = X^{(k)} + D^{-1}r_k$$

$A = D - U - L$ 取 $M = D - L \rightarrow$ 高斯-赛德尔迭代法

$$A = M - (M - A)$$

$$AX = b \rightarrow MX = (M - A)X + b$$

$$\rightarrow X^{(k+1)} = M^{-1}[(M - A)X^{(k)} + b]$$

$$\rightarrow X^{(k+1)} = X^{(k)} + M^{-1}[b - AX^{(k)}]$$

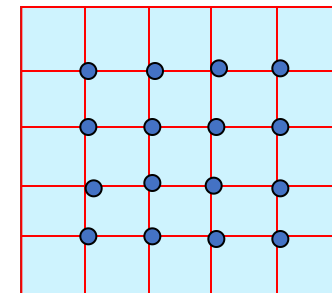
总结：简单迭代法

$$X^{(k+1)} = X^{(k)} + \omega(b - AX^{(k)})$$

迭代矩阵 $B = I - \omega A$

平面温度场问题:

$$\begin{cases} u_{xx} + u_{yy} = 0, & 0 < x, y < 1 \\ u(0, y) = u(x, 0) = u(x, 1) = 0 \\ u(1, y) = \sin \pi y \end{cases}$$



令 $h = 1/(n+1)$, $x_i = ih$, $y_j = jh$ ($i, j = 0, 1, \dots, n+1$)

记 $u_{i,j} = u(x_i, y_j)$, ($i, j = 0, 1, \dots, n+1$)

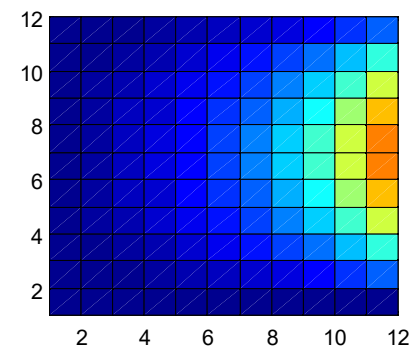
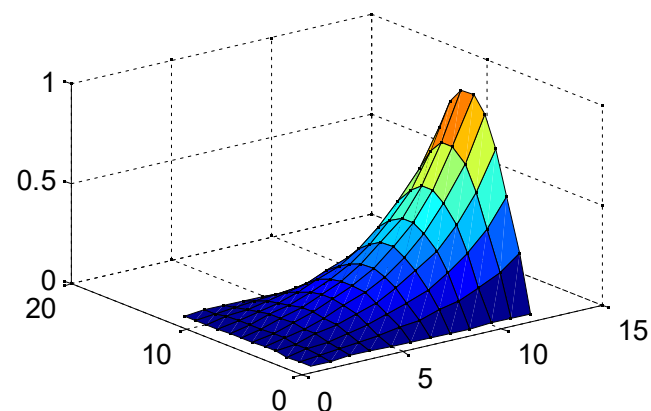
差分格式:

$$u_{i,j-1} + u_{i-1,j} - 4u_{ij} + u_{i+1,j} + u_{i,j+1} = 0$$

矩阵形式:

$$AU = F$$

$$A = \begin{bmatrix} B & I & & \\ I & B & I & \\ & I & B & I \\ & & I & B \end{bmatrix} \quad B = \begin{bmatrix} -4 & 1 & & \\ 1 & -4 & 1 & \\ & 1 & -4 & 1 \\ & & 1 & -4 \end{bmatrix}$$



高斯-赛德尔迭代法实验 (误差限 10^{-8}):

结点数 n^2	10^2	20^2	40^2
迭代次数	182	606	2077
CPU时间(s)	0.97	4.328	58.531
误差	0.0023	$6.4274e-4$	$1.6814e-4$

大型稀疏矩阵的背景

Jacobi迭代与Seidel迭代

迭代法的矩阵表示

迭代法数值实验